

Group Project: Business Analytics Research

Overview

The goal of this project is to give you hands-on experience in **formulating a business analytics research question**, identifying and preparing a **real-world dataset**, and applying **machine learning techniques in R** to provide insights or predictions. You will work in groups to develop your project, write a research paper in an academic style, and present your findings at the end of the semester.

Objectives

By completing this project, you will:

1. Learn how to identify and frame a research question in business analytics.
2. Gain experience with real-world datasets (from public repositories or self-collected).
3. Apply machine learning models in **R** for prediction and evaluation.
4. Develop skills in research writing and oral presentation.
5. Collaborate effectively as a team on a substantial analytics project.

Project Requirements

1. Research Question

- Define a clear research question relevant to business analytics.
- Your question should involve **prediction using machine learning**.
- Examples: *Predicting customer churn, forecasting sales, identifying fraudulent transactions, predicting employee attrition, etc.*

2. Dataset

You have two options for your dataset:

- **Public Dataset:** Use a dataset from repositories such as Kaggle, Google Dataset Search, UCI Machine Learning Repository, or government/industry open data portals.
- **Self-Collected Dataset:** You may gather your own data, for example:
 - **Web scraping/crawling** (e.g., product reviews, job postings, financial news).
 - **Surveys or questionnaires** (collecting primary data relevant to your research question).
 - **APIs or other data services** (where legally and ethically allowed).

Requirements:

- The dataset must be sufficiently large and feature-rich to support predictive modeling.
- You must document the source and method of collection.

3. Methodology

- Preprocess and clean the dataset.
- Select and justify your choice of machine learning model(s) in **R**. Python is not acceptable.
- Evaluate your model performance using appropriate metrics (e.g., accuracy, RMSE, AUC, F1-score).
- Discuss the business implications of your results.

4. Deliverables

(a) Research Paper

- Format: Follow the [CSWIM paper template](#).
- Length: 6 pages.
- Content:
 - Title, abstract, and keywords
 - Introduction
 - Literature review
 - Dataset description and preprocessing
 - Methodology
 - Results
 - Discussion and conclusion
 - References

(b) Code Appendix

- Submit all R scripts used for data preparation, modeling, and evaluation.
- Your code should be clean, well-commented, and reproducible.

(c) Presentation

- Each group will present their project at the end of the semester.
- Time: 15 minutes, followed by 5 minutes Q&A.
- Content: Research question, dataset, methods, key results, and business implications.

Grading

1. Group Grade (1–5 Scale)

- **1 (Incomplete / Errors)**
 - Coding errors prevent execution, OR the project is incomplete.
- **2 (Basic / Recycled Work)**
 - The research question is old, widely studied, or directly taken from class examples.
 - The dataset is commonly used in this or prior classes.
 - Uses only **one method** from class (e.g., regression).
- **3 (Standard / Novel Question & Two Methods)**

- Defines a **new research question**, supported by a literature review that shows novelty and interest.
- Uses a dataset (public or self-collected) that fits the question and is not one of the “standard class datasets.”
- Applies **at least two methods** (e.g., regression, decision trees).
- **4 (Advanced / Multiple Methods & Novelty)**
 - Meets all requirements of **3**.
 - Applies **five methods**, including at least **two new methods** developed within the last 10 years (e.g., deep learning models).
- **5 (Exceptional / Method Development)**
 - Meets all requirements of **4**.
 - Uses **five methods total**, including the **development of a new method with novelty** (e.g., revising the Transformer architecture to address a technical challenge).

2. Individual Grades

- Each student’s grade will start from the **group grade (1–5)**.
- Peer evaluations within the group will adjust individual grades to reflect relative contributions (see course peer evaluation policy).

3. Final Grade Algorithm

Step 1. Inputs

- **Group grade:** $G \in \{1,2,3,4,5\}$
- **Peer ratings:** Each member i in group j is rated by their teammates (scale 1–5).

Let:

- R_{ij} = average peer rating received by member i in group j .
- $\bar{R}_j$ = average of all R_{ij} in group j .

Step 2. Normalize Peer Ratings Within Group

We want to distribute the group grade while rewarding stronger contributions.

Define **peer weight**:

$$W_{ij} = \frac{R_{ij}}{\bar{R}_j}$$

- If $R_{ij} = \bar{R}_j$, then $W_{ij} = 1$ (average contributor).
- If above average, $W_{ij} > 1$.
- If below average, $W_{ij} < 1$.

Step 3. Scale Individual Grades

Now compute the individual's grade as:

$$I_{ij} = \min (5, G \times W_{ij})$$

- This ensures no one exceeds 5.
- Members who contributed more than average get a higher than the group grade.
- Those who contributed less get lower.

Example

- Group grade $G = 4$.
- Peer averages in group:
 - A: 4.8, B: 4.5, C: 4.2, D: 3.5, E: 3.0.
- Group mean $\bar{R}_j = 4.0$.
- Weights: A=1.20, B=1.13, C=1.05, D=0.88, E=0.75.
- Final grades (capped at 5):
 - A: 4.8, B: 4.5, C: 4.2, D: 3.5, E: 3.0.